Small learner pilot

Purpose: identify usability problems and collect initial evidence about the orbital lesson. This is a proposed protocol; no sessions or learning gains have been reported.

Session (10–15 minutes)

1. Ask permission to observe task behaviour and record anonymous notes. Use participant IDs rather than names. Explain that the interface is being tested, not the participant.

2. Ask: “A rocket has reached 200 km. What else, if anything, does it need to stay in orbit?” Record the explanation verbatim without coaching.

3. Ask the participant to find Mission Lab and record a prediction. Observe whether labels, controls, and speed units are understood.

4. Let them review and launch the showcase. Record assistance given; do not count an assisted step as unassisted completion.

5. Ask them to find periapsis on the map and decide whether the orbit clears the atmosphere.

6. Ask them to complete the mission, read the learning report, and save a reflection.

7. Ask the original concept question again, then a transfer question: “Would the required speed be the same around the Moon at the same altitude? Why?”

8. Ask what was confusing and what single change would help most.

Notes template

Field	Record
Participant ID	P01; no name or contact details
Date, device, browser	Actual session information
Before explanation	Verbatim response
Prediction recorded	Yes/no; assistance given
Launch completed	Yes/no; assistance given
Periapsis interpreted	Participant's words; correctness separately assessed
After explanation	Verbatim response
Moon transfer response	Verbatim response
Confusing controls or wording	Specific observation
Suggested change	Participant's words
Change implemented and retest	Evidence and actual date

Interpretation

Report the actual sample size, recruitment method, assistance, and failures. A small convenience sample is useful for finding defects, not for claiming general learning effectiveness. Separate preference feedback from conceptual understanding. Never replace missing sessions with simulated participants or generated testimonials.

A useful descriptive outcome is whether an explanation mentions sideways motion and continual gravitational falling after using the lesson. A rigorous learning claim would need a more carefully designed study, consistent assessment, and appropriate comparison conditions.